OLinuXino EEPROM content

A20 and T2 board-information data format

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Status: Maintained replacement specification

This document corrects and supersedes the July 2018 and October 2019 A20 EEPROM documents. It describes the 256-byte board-information record used by Olimex A20 and T2 OLinuXino products with a compatible EEPROM.

Implementation baseline: Olimex U-Boot branch `release-20250305`
The U-Boot source remains authoritative for the current board-ID registry.

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1. Scope

This specification defines the first 256 bytes of the board-information EEPROM used by supported Olimex boards based on the Allwinner A20 or T2 processor. It covers the A20-OLinuXino-LIME, LIME2, MICRO, A20-SOM, A20-SOM204 and their T2 industrial variants when the hardware contains a compatible EEPROM.

The board module owns the identification record. A carrier or evaluation board does not create a second board identity. For example, an A20-SOM204 mounted on an evaluation board is identified by the EEPROM on the SOM204.

> **Compatibility note.** Older hardware may have no EEPROM or may contain erased or invalid data. Software must validate both the magic value and CRC before using any field. A board name must be resolved from the current U-Boot board-ID registry, not guessed from configuration bytes alone.

1.1. Physical EEPROM addressing

The commonly used AT24C16-compatible EEPROM occupies I2C addresses `0x50` through `0x57`. The data structure in this document occupies the first 256-byte page, beginning at address `0x50`. Designers must consider all eight addresses when adding devices to the same I2C/TWI bus.

1.2. Normative implementation references

The implementation corresponding to this revision is found in the Olimex U-Boot repository:

- `include/olimex/sys_eeprom.h` - packed 256-byte structure.
- `board/olimex/common/sys_eeprom.c` - EEPROM I/O and CRC validation.
- `include/olimex/boards.h` - storage, grade and capacity encodings.
- `board/olimex/common/boards.c` - current board-ID registry.

Repository: https://github.com/OLIMEX/u-boot-olinuxino

2. General encoding rules

- The record length is exactly 256 bytes.
- All multi-byte integers are unsigned and stored little-endian.
- Text fields contain ASCII bytes and are not NUL-terminated unless explicitly stated.
- Capacity fields contain a base-2 exponent. A value n represents 2^n bytes.
- Erased EEPROM bytes normally read as `0xFF`.
- Readers must reject a record with an invalid magic value or CRC.

3. Memory layout

Offsets	Length	Field	Encoding
0x00-0x03	4	Header	Magic bytes 55 AA 4C 4F
0x04-0x07	4	Board ID	Unsigned 32-bit little-endian integer
0x08-0x09	2	Revision	Major and minor ASCII bytes
0x0A-0x0D	4	Serial number	Unsigned 32-bit little-endian integer
0x0E-0x11	4	Configuration	Storage, size, RAM and grade bytes
0x12-0x1D	12	Ethernet MAC	Twelve ASCII hexadecimal characters
0x1E-0xFB	222	Reserved	Reserved for future use
0xFC-0xFF	4	CRC-32	Unsigned 32-bit little-endian CRC

The MAC field ends at 0x1D. Offset 0x1E is the first reserved byte.

4. Field definitions

4.1. Header

The first four bytes are fixed:

Offset	0x00	0x01	0x02	0x03
Value	0x55	0xAA	0x4C	0x4F

Interpreted as a little-endian 32-bit integer, these bytes equal 0x4F4CAA55. A different value means the record is invalid.

4.2. Board ID

The board ID at offsets 0x04-0x07 is a unique unsigned 32-bit identifier stored little-endian. It selects the product name, expected storage, RAM, grade and device-tree configuration.

Board IDs are maintained in board/olimex/common/boards.c. Applications should update their lookup table from that source. Unknown IDs must be reported as unknown while the remaining structurally valid fields may still be shown.

4.3. Revision

Offset	Meaning	Valid value
0x08	Major revision	ASCII A through Z
0x09	Minor revision	ASCII 1 through 9, or unused

U-Boot writes 0x00 when the minor revision is unused. Some production or erased records contain 0xFF. Readers should display no minor revision for any byte outside ASCII 1 through 9.

Examples are C, G2 and J1.

4.4. Serial number

Offsets 0x0A-0x0D contain a sequential unsigned 32-bit serial number in little-endian order. U-Boot commonly displays it as eight hexadecimal digits.

4.5. Configuration

The configuration occupies four independent bytes:

Offset	Field	Meaning
0x0E	Storage	Primary onboard nonvolatile storage type
0x0F	Storage size	Capacity exponent, valid when storage is not 0x00
0x10	RAM size	Capacity exponent
0x11	Grade	Temperature-grade classification

4.5.1. Storage type

Value	ASCII	Meaning
0x00	-	No onboard storage represented by this field
0x65	e	eMMC
0x6E	n	NAND flash
0x73	s	SPI flash

The letters are lowercase. Uppercase E, N and S are not the encoding used by current Olimex U-Boot.

Some product names describe more than one fitted storage device, for example e16Gs16M. The configuration byte records the primary storage type; U-Boot identifies additional SPI flash from the board ID.

4.5.2. Capacity encoding

For both storage size and RAM size:

$$\text{capacity in bytes} = 2^{\text{field value}}$$

Value	Capacity	Value	Capacity
0	1 B	10	1 KiB
20	1 MiB	24	16 MiB
29	512 MiB	30	1 GiB
32	4 GiB	33	8 GiB
34	16 GiB		

Product names traditionally use MB and GB even though the encoding is based on powers of two.

4.5.3. Grade

Value	Meaning
`0x00`	Commercial grade
`0x01`	Industrial grade

4.6. Ethernet MAC address

Offsets `0x12-0x1D` contain exactly twelve uppercase ASCII hexadecimal characters, without separators. For example:

`301F9AD0ACB6` represents `30:1F:9A:D0:AC:B6`.

If a valid MAC address was not programmed, the field can contain twelve `F` characters or erased `0xFF` bytes. Readers should display `FF:FF:FF:FF:FF:FF` and must not install it as an Ethernet address.

Original A20-SOM products do not contain an onboard Ethernet PHY and normally do not carry a usable Ethernet MAC. A20-SOM204 behavior is variant-dependent. In the current U-Boot registry, IDs 8958, 10257 and 10234 are classified as having EMAC. Software should use the current board-ID registry rather than assuming that every SOM204 variant has or lacks Ethernet.

4.7. Reserved area

Offsets `0x1E-0xFB` are reserved. Current U-Boot initializes this area to `0xFF`. Readers must ignore it. Writers should initialize it to `0xFF` for a new record and should not assign private meanings that could conflict with future revisions.

4.8. CRC-32

Offsets `0xFC-0xFF` contain the CRC-32 of bytes `0x00-0xFB`, stored as a little-endian unsigned 32-bit value.

The algorithm is the U-Boot `crc32()` implementation invoked as:

```
crc = crc32(0L, record, 252);
```

The CRC field itself is excluded from the calculation. A record is valid only when the stored value equals the calculated value.

5. Validation procedure

A reader should perform these steps in order:

1. Read the full 256-byte page beginning at I2C address `0x50`.
2. Confirm bytes 0-3 are `55 AA 4C 4F`.
3. Calculate CRC-32 over bytes 0-251 with initial value zero.
4. Decode the stored CRC from bytes 252-255 as little-endian.
5. Reject the record if the CRC values differ.
6. Decode the remaining fields.
7. Resolve the board ID using the current board registry.

A diagnostic tool may offer an explicit option to decode a record with a bad or unavailable CRC, but it must clearly mark the result as unverified.

6. Worked configuration examples

Product configuration	Storage byte	Size byte	RAM byte	Grade byte
No storage, 1 GiB RAM, commercial	`0x00`	Not applicable	30 (`0x1E`)	`0x00`
4 GiB eMMC, 1 GiB RAM, commercial	`0x65`	32 (`0x20`)	30 (`0x1E`)	`0x00`
16 GiB eMMC, 1 GiB RAM, commercial	`0x65`	34 (`0x22`)	30 (`0x1E`)	`0x00`
16 MiB SPI, 512 MiB RAM, industrial	`0x73`	24 (`0x18`)	29 (`0x1D`)	`0x01`

7. A20-SOM and A20-SOM204 board IDs

This table is an informative snapshot from Olimex U-Boot `release-20250305`. The source registry is authoritative.

ID	Board name	Storage	Grade
4673	A20-SOM-n4GB	NAND 4 GiB	Commercial
7664	A20-SOM	None	Commercial
8849	A20-SOM-IND	None	Industrial
8922	A20-SOM-n8GB	NAND 8 GiB	Commercial
9155	A20-SOM-e16GB	eMMC 16 GiB	Commercial
9148	A20-SOM-e16GB-IND	eMMC 16 GiB	Industrial
9047	A20-SOM-e16Gs16M	eMMC 16 GiB	Commercial
9259	T2-SOM-IND	None	Industrial
9827	T2-SOM-e8Gs16M-IND	eMMC 8 GiB	Industrial
11454	T2-SOM-e16Gs16M-IND	eMMC 16 GiB	Industrial
8991	A20-SOM204-1G	None	Commercial
8958	A20-SOM204-1Gs16Me16G-MC	eMMC 16 GiB	Commercial
10257	A20-SOM204-1G-M	None	Commercial
10157	T2-SOM204-1Gs16Me4G-C-I	eMMC 4 GiB	Industrial
10234	T2-SOM204-1Gs16Me8G-MC-I	eMMC 8 GiB	Industrial
11458	T2-SOM204-1Gs16Me16G-M-I	eMMC 16 GiB	Industrial
11462	T2-SOM204-1Gs16Me16G-MC-I	eMMC 16 GiB	Industrial
10238	T2-SOM204-1G-I	None	Industrial

8. Change-control guidance

Future changes should commit both this LaTeX source and its generated PDF. Changes to the binary layout require a corresponding implementation change and a new document release date. Updates to product IDs normally belong only in `boards.c`; this document should include a dated snapshot only when it is useful for users.